Unit: Unit 0: Foundations of Algebra
Topic: Natural Numbers Integers

Name: ____________

Date: ________

Challenge (Habanero)

Q1. A hiker ascends 5 mountains, each with an elevation of -200 feet below sea level. What is the total elevation change?

(A) -1000 feet
(B) 1000 feet
(C) 0 feet
(D) -500 feet
(E) 500 feet

Q2. A park ranger tracks a pack of wolves, observing their population grows by 3 every 2 months. If the pack starts with 17 wolves, how many will there be after 8 months?

(A) 23 wolves
(B) 25 wolves
(C) 29 wolves
(D) 31 wolves
(E) 35 wolves

Q3. A gas tank is filled with 25 gallons of fuel. If 3 gallons are consumed every 50 miles, how many miles can the vehicle travel on -5 gallons of fuel?

(A) The vehicle cannot travel
(B) -80 miles
(C) -50 miles
(D) 0 miles
(E) 50 miles

Q4. What is the next number in the pattern: $2, 5, 8, 11, 14, ...$?

(A) 17
(B) 20
(C) 21
(D) 23
(E) 25

Q5. A group of friends plan a trip to a national park, and they drive a total distance of 240 miles. If they drive at an average speed of -8 miles per hour, what is their total travel time?

(A) The trip is not possible
(B) -30 hours
(C) -20 hours
(D) 0 hours
(E) 30 hours

Q6. What is the result of $(-3) \cdot (-5)$?

(A) -15
(B) 15
(C) -9
(D) 9
(E) 0

Q7. A camper observes that the temperature drops by 2 degrees every hour. If the initial temperature is -5 degrees, what is the temperature after 4 hours?

(A) -13 degrees
(B) -9 degrees
(C) -5 degrees
(D) -1 degree
(E) 1 degree

Q8. A hiker needs to ascend a mountain with an elevation change of 1200 feet. If the hiker climbs 4 hours a day at a rate of -3 feet per hour, how many days will it take to reach the top?

(A) The hiker will never reach the top
(B) -100 days
(C) 100 days
(D) 50 days
(E) 40 days

Open Ended Questions (FRQ)

Q9. A group of friends plan a road trip, and they want to know how much fuel they will consume. If their vehicle's fuel efficiency is 25 miles per gallon, and the total distance of the trip is 1200 miles, how much fuel will they consume if they start with a full tank of 20 gallons?

Q10. A wildlife reserve tracks the population of a certain species of animal. If the population grows by 15 every 3 months, and the initial population is 50, what is the population after 12 months?

Chef's Tips

- Tip 1: Ensure students understand the difference between natural, whole, and integer numbers.
- Tip 2: Emphasize the importance of signs when performing operations with integers.
- Tip 3: Encourage students to visualize the problems and use real-world examples to illustrate the concepts.